BLOCKCHAIN FORENSICS

INVESTIGATIVE TRACE REPORT

Case Study 5: 94-Day Multi-Layer Bitcoin Laundering Operation

February 15 — May 20, 2020 | $390,284+ Total Operation Value

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Tools: WalletExplorer.com · mempool.space · Chainabuse · Custom Python Red-Flag Scorer v3

Executive Summary

This report documents a complete end-to-end trace of a sophisticated Bitcoin money laundering operation spanning 94 days, from February 15 to May 20, 2020. The investigation began with a single collection wallet moving $3,819 in structured Bitcoin deposits and expanded through systematic chain tracing to reveal a multi-layer, multi-track laundering architecture with a total operation value exceeding $390,284.

The trace was conducted entirely using open-source blockchain intelligence tools and a custom Python red-flag scoring framework developed by the analyst. No paid attribution services were used. The operation demonstrates advanced laundering tradecraft including automated HD wallet generation, time-staggered consolidation, deliberate dormancy periods, round-number output engineering, and dual cash-out at two separate high-volume exchange clusters.

The investigation identified two parallel laundering tracks originating from the same source infrastructure, both ultimately terminating at massive exchange clusters — one with 3.9 million transactions and one with over 55 million transactions — consistent with major international cryptocurrency exchanges where KYC records and transaction data would be available through legal process.

Operation Overview

Wallet ID	Role	Duration	BTC Value
5647dc464e	Collection Wallet	20 minutes	0.67675 BTC
2b6cab0a70	Splitter Wallet	13 minutes	0.67658 BTC
b013e7103a	Primary Track Wallet	35 minutes	0.60597 BTC
2fbebd61e3	Structured Aggregator	~6 hours	0.40866 BTC
8ac8508bc0	Mid-Level Aggregator	2 days	14.30 BTC
377788adc8	Pass-Through	3 hours	3.12830 BTC
00950b5318	46-Address Fan-Out	Single block	3.12830 BTC
3PXBsTb3g7...	Accumulation Wallet	13 day dormancy	15.44 BTC
a547750c33	Integration Wallet	39 day dormancy	7.17307 BTC
1c09ea1202	Final Consolidation	19 day dormancy	16.62557 BTC
0000011bd9	Cash-Out Exchange A	55M+ transactions	16.20 BTC
00002c1433	Cash-Out Exchange B	3.9M transactions	0.40866 BTC

Operation Timeline

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FEB 15, 2020  — First detected transaction — bc1q address receives dust
FEB 25, 2020  — Main operation launches
  10:18:57    — Three simultaneous deposits hit [5647dc464e] — automated
  10:38:27    — Collection wallet sweeps to splitter [2b6cab0a70]
  10:51:24    — Splitter divides: 89.6% → fast track / 10.4% → aggregator
  10:49–16:09 — Six structured deposits into [2fbebd61e3] over 6 hours
  16:49:04    — Aggregator sweeps $3,819 → Exchange Cluster A [00002c1433]
               ← TRACK 1 COMPLETE

MAR 07, 2020  — Accumulation wallet [3PXBsTb3g7] receives 8.04 + 7.40 BTC
               simultaneously in two transactions confirmed same second
               13 DAY DORMANCY BEGINS

MAR 20, 2020  — Dormancy ends — both batches move within 7 hours
  03:09:51    — 8.04 BTC splits → [1KbYXafo] $24,981 + [3KoU2H41] $43,364
  10:47:37    — 7.40 BTC splits → [3BsE1Yqv] $1,400 peel + [3DuP3JaA] $44,472
               39 DAY DORMANCY BEGINS

APR 28, 2020  — [a547750c33] sweeps 7.16 BTC → [1c09ea1202]
APR 28–MAY 1  — Four additional sources feed [1c09ea1202] over 3 days
               Total accumulated: 16.62557 BTC
               19 DAY DORMANCY BEGINS

MAY 20, 2020  — Final sweep: exactly 16.20 BTC → [0000011bd9]
               ← TRACK 2 COMPLETE — 55M+ transaction exchange cluster
               OPERATION COMPLETE — 94 DAYS TOTAL
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Chain Visualization

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[Automated sources — simultaneous same-second deposits]
                    ↓
          [5647dc464e] Collection — 20 min
                    ↓
          [2b6cab0a70] Splitter — 13 min
           ↙ 89.6%             10.4% ↘
[b013e7103a]                  [2fbebd61e3]
FAST TRACK — 35 min           STRUCTURED AGGREGATOR
  ↙ 58.6%   41.4% ↘           6 hrs · 6 feeders · CV=0.09
[8ac8508bc0] [0a74b1488d]              ↓
$133k · 2 days         [00002c1433] EXCHANGE A
     ↓                  3.9M transactions
[377788adc8]           ← TRACK 1 CASH OUT — $3,819
$29,242 · 3 hrs
     ↓
[00950b5318] 46 simultaneous child addresses — single block
     ↓
[3PXBsTb3g7...] Accumulation — $100k+ — 13 day dormancy
     ↓
[a547750c33] Integration — 39 day dormancy
     ↓
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[1c09ea1202] Final consolidation — 5 feeders — 19 day dormancy
             16.20 BTC ROUND NUMBER engineered
     ↓
[0000011bd9] EXCHANGE B — 55M+ transactions
← TRACK 2 CASH OUT — $150,660+
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Key Investigative Findings

Finding 1 — Automated Infrastructure Confirmed

Three deposits into the collection wallet [5647dc464e] arrived at the exact same second (10:18:57 UTC on February 25, 2020). Two separate transactions into the accumulation wallet [3PXBsTb3g7] confirmed at the exact same second on March 7, 2020. Simultaneous same-second blockchain confirmations from multiple source addresses are technically impossible through manual operation — this is scripted, programmatic infrastructure.

> **Significance:** The operation was not run by an individual manually moving funds. It was operated by automated software capable of coordinating multiple wallets simultaneously.

Finding 2 — HD Wallet Generation Confirmed

Wallet [00950b5318] distributed funds to exactly 46 Bitcoin addresses, all confirmed in the same block (620678). The 46 addresses include both legacy (3-prefix) and SegWit (bc1q-prefix) formats — confirming scripted multi-format address generation from a single seed phrase. Every address received exactly one transaction, drained to zero, and was never used again.

> **Significance:** The operator generated 46 disposable child wallets from a single HD wallet seed phrase, funded them simultaneously, and discarded them — classic automated layering infrastructure.

Finding 3 — Deliberate Structuring Under Reporting Thresholds

Six deposits into the structured aggregator wallet [2fbebd61e3] between 10:49 and 16:09 UTC on February 25, 2020 show a coefficient of variation of approximately 0.09 — well below the 0.15 threshold flagged by the analyst's red-flag scoring model as indicative of deliberate amount clustering. Individual deposit values ranged from $550.17 to $706.02.

- All six deposits fell below the $3,000 MSB record-keeping threshold (31 CFR §1022.410)
- All six deposits fell below the $10,000 CTR reporting threshold (31 USC §5313)
- The total sweep of $3,819 crossed the $3,000 MSB threshold at the receiving exchange

> **Statutory Exposure:** 31 USC §5324 — Structuring to evade reporting requirements. Prosecution does not require proving the underlying crime — the structuring pattern itself is the offense.

Finding 4 — Three Deliberate Dormancy Periods

The operation employed three distinct dormancy periods of increasing length as funds moved toward the cash-out point:

- 13 days — accumulation wallet [3PXBsTb3g7] — March 7 to March 20
- 39 days — integration wallet [a547750c33] — March 20 to April 28
- 19 days — final consolidation [1c09ea1202] — May 1 to May 20

Total dormancy across the operation: 71 of 94 days. The funds were actively moving for only 23 days out of the full 94-day operation window.

Significance: Progressive dormancy periods are a deliberate anti-detection technique — allowing blockchain analytics alerts to expire and investigator attention to move elsewhere before the next hop.

Finding 5 — Round Number Output Engineering

The final consolidation wallet [1c09ea1202] received 16.62557281 BTC from five sources and swept exactly 16.20000000 BTC to the cash-out exchange — a perfectly round number. Round BTC amounts do not occur naturally due to fee calculations and exchange rate conversions. The operator deliberately structured the five input amounts to produce a specific round output, indicating sophisticated operational coordination and pre-planned cash-out parameters.

Significance: Round number engineering indicates the cash-out destination specified an exact amount — consistent with an OTC desk, exchange account with a pre-arranged trade, or a second operator receiving a specific agreed sum.

Finding 6 — Dual Cash-Out at Two Separate Exchange Clusters

The operation terminated at two separate high-volume exchange clusters:

• Exchange Cluster A [00002c1433] — 3,934,605 transactions across 876,843 addresses — received Track 1 funds ($3,819) on February 25, 2020

• Exchange Cluster B [0000011bd9] — 55,000,000+ transactions — received Track 2 funds (16.20 BTC = $150,660+) on May 20, 2020

Both clusters are consistent with major international cryptocurrency exchanges. Definitive attribution requires paid blockchain analytics tools (Chainalysis Reactor, TRM Labs) or legal process.

Investigative Action: Subpoena Duces Tecum to exchange operators for KYC records, transaction logs, IP addresses, and device fingerprints associated with the receiving accounts.

Financial Summary

Wallet	BTC Amount	USD Value	Date
Collection [5647dc464e]	0.67675 BTC	$6,324.52	Feb 25, 2020
Structured aggregator sweep	0.40866 BTC	$3,818.90	Feb 25, 2020
Mid-level aggregator [8ac8508bc0]	14.30348 BTC	$133,683.50	Feb–Mar 2020
Accumulation wallet total	15.44409 BTC	$100,000–$139,000	Mar 2020
March 20 total movement	15.40 BTC	$114,217.00	Mar 20, 2020
Final consolidation total	16.62557 BTC	$129,679.47	Apr–May 2020
Final cash-out sweep	16.20000 BTC	$150,660.00	May 20, 2020
TOTAL OPERATION VALUE	~41.84 BTC	$390,284+	Feb–May 2020

Applicable Federal Statutes

Money Laundering — Laundering of monetary instruments. The layered wallet structure, dormancy periods, and dual cash-out demonstrate knowing concealment of criminally derived proceeds. — 18 USC §1956

Structuring — Deliberate division of funds into sub-threshold amounts to evade CTR and SAR reporting requirements. CV analysis confirms non-random clustering. — 31 USC §5324

Wire Fraud — Automated scripted transactions using wire communications in furtherance of a scheme to defraud. — 18 USC §1343

Bank Secrecy Act compliance obligations — Exchange clusters receiving funds are MSBs subject to KYC, CTR, and SAR requirements. Records are legally required to exist. — 31 USC §5318

Recommended Investigative Next Steps

• Submit wallet cluster addresses [00002c1433] and [0000011bd9] to Chainalysis Reactor or TRM Labs for exchange attribution — scale strongly suggests Binance, Huobi, or OKX

• Issue Subpoena Duces Tecum to identified exchanges for KYC records, IP logs, device fingerprints, and fiat withdrawal bank account information associated with receiving accounts

• Trace remaining unresolved destinations — [3KoU2H41Yj6szeEydZYX4QKseLhyKtpFLU] ($43,364) and [1KbYXafoZ1RwKK5iMZgvoHyWnpRRDHSLGT] ($24,981) — both March 20 outputs not yet fully traced

• Trace the outside input [3Bkwz7TWidYKxWhYVnuxVecK7sf2dMNztE] that contributed 2.99 BTC to the March 20 transaction — this wallet fed in from outside our traced chain and may represent a parallel operation using the same cash-out infrastructure

• Run all six feeder wallets into [2fbebd61e3] through the red-flag scorer and trace upstream — shared parent wallet would confirm single coordinated operation

• Cross-reference February 25, 2020 and May 20, 2020 against known criminal investigations, exchange hacks, darknet market activity, or ransomware payments for contextual attribution

• Analyze the 11 peel addresses from the March 7 transaction receiving 0.038–0.045 BTC each — tight clustering suggests additional structuring layer worth documenting

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Methodology Note

This trace was conducted using the following open-source tools and self-developed analytical frameworks:

• WalletExplorer.com — wallet clustering, transaction history, named service identification

• mempool.space — raw transaction data, UTXO analysis, block confirmation verification, visual transaction mapping

• Custom Python Red-Flag Scorer v3 — seven-indicator statistical model including drain-to-zero detection, CV-based clustering analysis, value-weighted sweep dominance, dormancy gap measurement, and organic-usage dampener

• Portnoff Timestamp Methodology — adapted from UC Berkeley KDD 2017 research for correlating transaction timing with operational patterns

All findings are based on publicly available blockchain data. No proprietary tools, paid services, or non-public information were used in this analysis. The analytical conclusions represent the analyst's independent interpretation of on-chain evidence and should be cross-referenced against proprietary attribution databases before being used as the basis for legal action.

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This document is an original analytical work product. All wallet addresses and transaction data are sourced from the public Bitcoin blockchain.